

Vibration control of vortex-induced vibrations of a bridge deck by a single-side pounding tuned mass damper

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ABSTRACT

This paper proposes a new method to mitigate vortex-induced vibrations (VIVs) of a bridge deck using a single-side pounding tuned mass damper (SS-PTMD). The SS-PTMD is a passive control device and comprises an undamped tuned mass with a pounding boundary covered with viscoelastic (VE) materials layer. A nonlinear force model for describing impact behavior of VE materials is used to simulate the response of a single degree of freedom (SDOF) system controlled by a SS-PTMD. The free pounding experiments are performed to determine the model parameters of impact force and validate the simulation method. The optimal design of SS-PTMD for SDOF system subjected to sinusoidal excitation is carried out by numerical optimization, and the optimized SS-PTMD is applied to control the vertical VIVs of a bridge deck. The control performance is experimentally examined by elastically mounted section model tests in wind tunnel. In addition, the classical wake oscillator model is used to predict the behavior of the coupled fluid-structure system under VIV and explore the control performance of the SS-PTMD. The experimental results show that the maximum response of the bridge deck model was reduced by 94% when a SS-PTMD with mass ratio of 2% was applied. It is also shown that non-linearity in vortex shedding forces has little influence on control performance of SS-PTMD optimized under sinusoidal excitation.

1. Introduction

Vortex-induced vibrations (VIVs) of the bridge deck happens if the frequency of vortex-shedding matches with one of the natural frequencies of the bridge [1–4]. It is a resonant-type vibration dominated by a single mode of vibration, and has been observed on many bridges. For the long-span bridges with small damping and more flexible configurations, large amplitude VIV may occur in a relatively low wind speed range about 5–10 m/s. In another word, long-span bridges may suffer from long-term vibration with considerably large amplitudes, which discomforts users and greatly shorten the life of bridges.

Tuned mass dampers (TMDs), as a low-cost and simple passive control device, have been applied to vibration control of flexible and low-damping structures under different excitation inputs, such as vehicle [5–8], wind [9–12] and earthquake excitations [13–16]. The effectiveness of using TMD to control VIVs of flexible structures, such as pipelines [17,18], high-rise buildings [19–21] and long-span bridges [22,23], were reported. The conventional TMD, which is firstly proposed by Frahm [24], only consists of a mass-spring system. When the

frequency of the mass is exactly tuned to the frequency of the controlled structure, the vibration energy of the controlled structure will be transferred to the tuned mass. Den Hartog incorporated a linear viscous damper into TMD and the optimal parameters of TMDs to control an undamped system was derived [25,26]. Some other solutions to supply linear or nonlinear damping for TMDs were studied, such as particle dampers [27–29], Magneto-rheological (MR) dampers [30–33], friction dampers and eddy current dampers [34–38].

Recently, Song et al. [39] introduced the pounding damping into a TMD system to enhance the control performance and the combined system is called pounding tuned mass damper (PTMD), as shown in Fig. 1(a). In their design, two pounding boundaries are added in the either sides of the tuned mass and this type of damper is termed as a double-side PTMD. When the relative vibration displacement exceeds the gap between the tuned mass and the pounding boundaries, the tuned mass will pound on the boundary to provide supplemental energy dissipation. To enhance the energy dissipation during poundings, viscoelastic (VE) material layer is attached on the pounding boundary. Several applications of the double-side PTMD for vibration control of a

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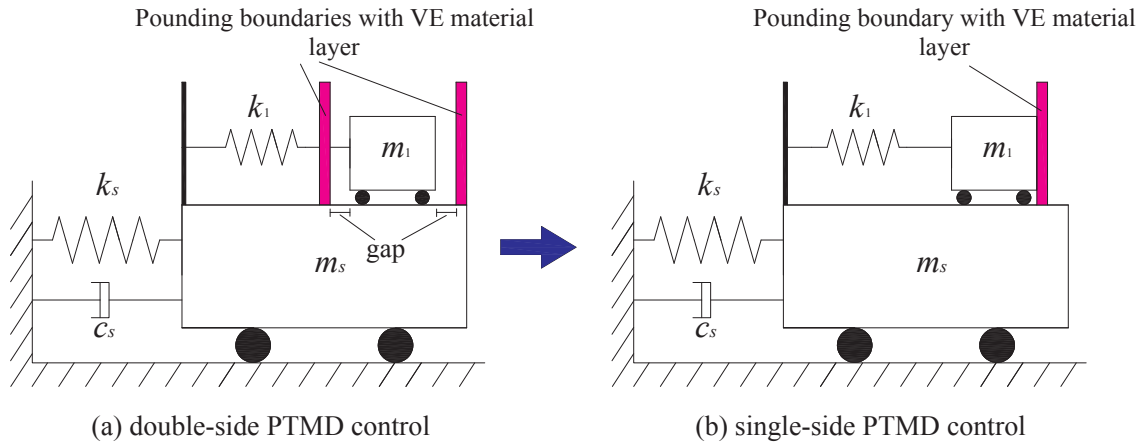


Fig. 1. A SDOF system with conventional PTMD and SS-PTMD.

traffic pole [40,42] and subsea pipelines [39,41,43] showed that the double-side PTMD is more robust and effective than the conventional TMD [42–44]. Furthermore, the effectiveness of the double-side PTMD for the seismic vibration protection of multiple degree of freedom (MDOF) structures was also investigated [45,46]. Based on the double-side PTMD design, Wang et al. [47,48] proposed a novel single-side pounding tuned mass damper (SS-PTMD) to control the free vibration and forced vibration of a low-damping structure, as shown in Fig. 1(b). Compared to the conventional double-side PTMD, only one pounding boundary is added next to the equilibrium position of the spring-mass system to provide additional damping for the TMD system. This modification eliminates the gap between the mass and pounding boundary as shown in Fig. 1(a) and simplifies the parametric design. The optimal design of the SS-PTMD for controlling a SDOF system subjected to sinusoidal excitation has been studied [49,50].

In many situations, the installation space and stroke of tuned mass dampers is a practical constraint and the SS-PTMD provides a superior solution. Moreover, the energy dissipation provided by VE materials during poundings is not changed too much even after 350,000 poundings [51]. Although the effectiveness of the novel SS-PTMD to control the free vibration and the forced vibration under harmonic loadings were demonstrated, application to control the VIVs of bridge girders is still an interesting topic and has not been reported yet. Unlike sinusoidal excitation, vortex shedding forces responsible for VIV of bridge deck involves structure-fluid interaction and is highly nonlinear. In this study, the effectiveness of SS-PTMD for control of VIVs of a bridge deck is investigated experimentally and numerically. An impact force model which considers the nonlinear impact behavior of VE materials is used to simulate the response of a single degree of freedom (SDOF) system equipped with a SS-PTMD. The optimal parameters of the SS-PTMD are obtained for a SDOF structure subjected to sinusoidal excitations. Moreover, the optimized SS-PTMD is applied to control the vertical VIV of a bridge girder sectional model. The effectiveness of the proposed SS-PTMD to control VIVs is numerically studied by the wake oscillator model and examined by the wind tunnel experiments.

2. Modeling of single-side pounding tuned mass damper (SS-PTMD)

2.1. Dynamic properties of SS-PTMD

The mechanical model of a single degree of freedom (SDOF) structure equipped with the novel SS-PTMD to suppress vertical vibrations is shown in Fig. 2. The SS-PTMD has a pounding boundary covered by VE material layers beneath the tuned mass when the mass is in the equilibrium position. The natural frequency f_d and the equivalent damping ratio ζ_d of the novel SS-PTMD can be expressed as [48],

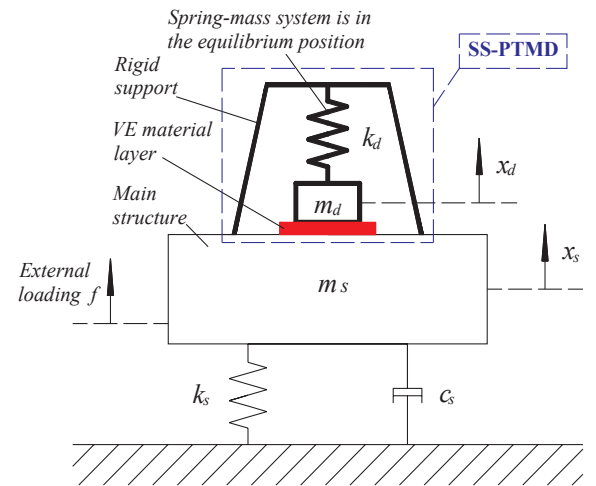


Fig. 2. A vertical vibration system controlled by the proposed SS-PTMD.

$$f_d = \frac{1}{\pi} \sqrt{\frac{k_d}{m_d}} \quad (1)$$

$$\zeta_d = \frac{1}{2\pi} \ln\left(\frac{1}{e}\right) \quad (2)$$

where k_d and m_d are the stiffness and mass of the SS-PTMD, e is the coefficient of restitution (COR) during impacts between tuned mass and the VE material layer. Contrast to a general spring-mass system, it is evident that the frequency of the SS-PTMD is doubled. This is because the pounding boundary prevents the mass move into the lower half stroke, as shown in Fig. 2. Since Eq. (1) ignores the pounding duration, the actual frequency should be slightly larger than that provided by Eq. (1). Moreover, it should be noted that Eq. (2) is an approximate expression for damping ratio calculation and it has good accuracy when e is larger than 0.4 [47].

2.2. Pounding force modeling

The conventional impact theory, employing the impact velocity exchange, neglects the pounding duration and cannot provide the specific value of the impact force [49]. In this paper, an impact force model, which considers the nonlinear behavior of poundings between VE material layer and steel, is utilized to simulate the response of the SS-PTMD. The pounding force between the VE materials layer and steel is given by [49],

$$F_p(t) = \begin{cases} k\delta^n(t) + \zeta\dot{\delta}^n(t)\dot{\delta}(t) & \delta_{\max} \geq \delta(t) > 0, \quad (\text{approaching period}) \\ & \dot{\delta}(t) \geq 0 \\ f_e \left[\frac{\delta(t) - e_1\delta_{\max}}{(1-e_1)\delta_{\max}} \right]^n & \delta_{\max} > \delta(t) \geq \delta_e, \quad (\text{restitution period}) \\ & \dot{\delta}(t) < 0 \\ 0 & \delta_e > \delta(t) > 0, \\ & \dot{\delta}(t) < 0 \end{cases} \quad (3)$$

where k and ζ are the pounding stiffness and damping, respectively. $\delta(t)$ and $\dot{\delta}(t)$ are the indentation and relative velocity of the colliding bodies during pounding. n is the pounding nonlinear coefficient, $\delta_{\max}$ is the maximum indentation during pounding. e_1 and f_e are the remaining surface deformation ratio and the maximum elastic impact force, they can be expressed as [49],

$$e_1 = \frac{\delta_e}{\delta_{\max}} \quad (4)$$

$$f_e = k\delta_{\max}^n \quad (5)$$

where δ_e is the remaining surface deformation of the VE materials layer when the colliding bodies are just separated.

In the impact force model, the pounding stiffness k is determined by letting the maximum simulated impact force be equal to the measured value from free pounding experiments. The remaining surface deformation ratio e_1 also can be obtained from the pounding experiments by Eq. (4). The maximum indentation $\delta_{\max}$ can be collected during the pounding simulation. The pounding damping ζ , which represents the energy loss during one pounding, can be expressed as the following explicit formula [49],

$$\frac{1}{2}(1-e^2)\dot{\delta}_0^2 = (1-e_1) \left(\frac{k^2}{\zeta^2} \ln \left| \frac{\dot{\delta}_0 + \frac{k}{\zeta}}{-\frac{k}{\zeta}} \right| - \frac{k}{\zeta} \dot{\delta}_0 \right) + \frac{\dot{\delta}_0^2}{2} \quad (6)$$

where $\dot{\delta}_0$ is the relative pre-impact velocity of the colliding bodies. It should be noted that Eq. (6) can be solved by Newton iterative method to obtain the pounding damping for each pounding.

2.3. Free pounding experiment

In Eq. (2), the equivalent damping ratio of the SS-PTMD is only governed by the coefficient of restitution e . Therefore, a free pounding experiment was performed to study the coefficient of restitution of different VE materials under various pre-impact velocities, as shown in Fig. 3. In the experimental setup, a cantilever beam with a mass block at its end was fixed by a rigid support. The VE materials layer was placed upon a force sensor. The spherical striker was just contacted to the VE materials layer when the cantilever beam-mass system was in the equilibrium position. A laser displacement sensor (optoNCDT14,

manufactured by $\mu\epsilon$ Company) was used to measure the vertical displacement of the mass block, and the force sensor (PWSK-300N, manufactured by Shanghai Instruments Company) was used to collect the impact force during poundings. In each test, an initial vertical displacement x_{in} was given to the mass block and then let it freely pound on the VE materials layer. Based on the law of energy conservation, the pre-impact velocity V_{pre} and post-impact velocity V_{post} are given by,

$$V_{pre} = \sqrt{\frac{k_1}{m_1} x_{in}} \quad (7)$$

$$V_{post} = \sqrt{\frac{k_1}{m_1} x_{re}} \quad (8)$$

where x_{in} and x_{re} are respectively the initially released displacement and the maximum rebounded displacement after one pounding. m_1 is the weight of the mass block and k_1 is the stiffness provided by the cantilever beam which can be easily obtained from frequency test. Therefore, the coefficient of restitution e can be obtained by,

$$e = \frac{V_{post}}{V_{pre}} = \frac{x_{re}}{x_{in}} \quad (9)$$

The pre-impact velocity can be alternated by changing the initially released displacement. Two VE materials, as shown in Fig. 4, high energy dissipation rubber (HEDR, which was manufactured by Zhuzhou Times New Materials Technology Company) and very high bond (VHB, which was manufactured by 3M Company) were used in the experiment. Three different VE material layers were divided, namely 12 mm VHB layer, 12 mm HEDR layer and the composite layer composed of 3 mm VHB and 9 mm HEDR layer. In each case the different VE material layers with the same thickness of 12 mm was tested.

The relationship between the pre-impact velocity and the coefficient of restitution is shown in Fig. 5. It can be seen that the coefficient of restitution of each case is almost a constant under various pre-impact velocities. The average coefficients of restitution of the VHB materials and the HEDR are 0.3694 and 0.6296, respectively. The average coefficient of restitution of the composite layer composed of 3 mm VHB and 9 mm HEDR is 0.5008, which is somewhere in between the mean value of the case of VHB layer and the case of HEDR layer. This phenomenon indicates that the conception of using composite VE material layer can realize the adjustability of the equivalent damping ratio of the SS-PTMD in a certain range.

2.4. Validation of pounding force modeling

When a initially released displacement $x_{10} = 0.04$ m is given to the mass block and the 12 mm DEDR pounding layer is used in the free pounding experiment, the history of vertical displacement of the mass block and the impact force during poundings are shown in Fig. 5. To verify the accuracy of the impact force model, a simulation of the free

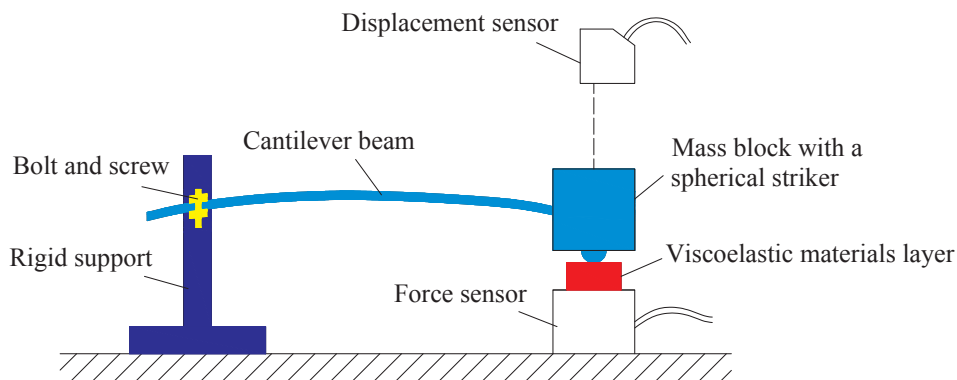
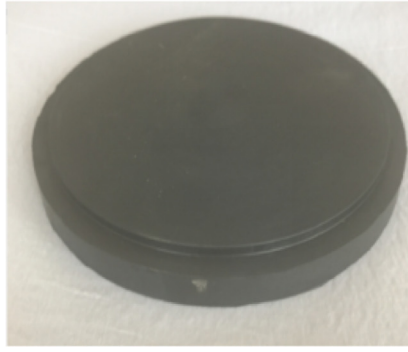


Fig. 3. Setup of the free pounding experiment.



(a) HEDR



(b) VHB

Fig. 4. Two kinds of VE materials.

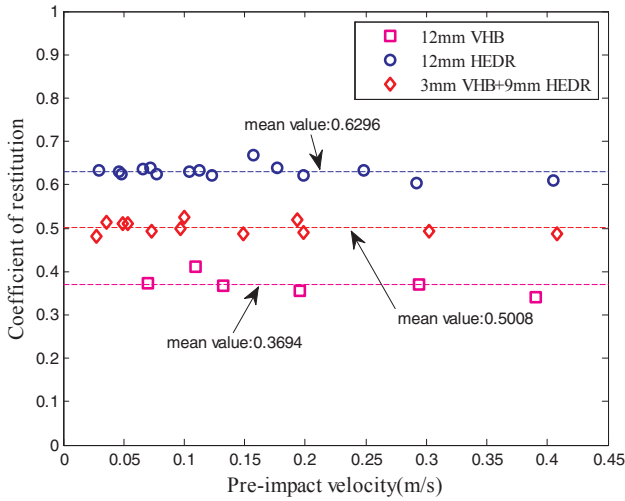


Fig. 5. Relationship between coefficient of restitution and different pre-impact velocities.

pounding experiment is carried out to compare the history of the displacement and impact force measured from the experiment. For the free pounding experiment, as shown in Fig. 3, the governing equation of motion is,

$$m_1 \ddot{x}_1(t) + k_1 x_1(t) = F_p(t) \quad (10)$$

where $\ddot{x}_1(t)$ and $x_1(t)$ are the vertical acceleration and displacement of the mass block, respectively. m_1 is the weight of the mass block, and k_1 is stiffness provided by the cantilever beam. $F_p(t)$ is the pounding force and it remains zero when $x_1(t) < 0$. $F_p(t)$ can be computed by substituting Eq. (3) into Eq. (10) when $x_1(t) > 0$ and the indentation $\delta(t)$ in Eq. (3) should be defined as,

$$\delta(t) = x_1(t) \quad (11)$$

According to the previous study on the pounding property of VE materials, the nonlinear pounding coefficient n is set to 1.5 [48] and the remaining surface deformation ratio e_1 is set to 0.5 [49]. The pounding stiffness is determined to be $6.4 \times 10^5 \text{ N/m}^{1.5}$ by letting the maximum force value 67.7 N of the first pounding from experiment be equal to that from simulation [52,53]. The Runge-Kutta fourth-order method was employed to solve Eq. (10) and the time interval was $1 \times 10^{-4} \text{ s}$. The initial condition of the simulation was set to $x_1 = 0.04 \text{ m}$ and $\dot{x}_1 = 0$. The weight of mass block is 0.86 kg and the stiffness provided by the cantilever beam is 85.5 N/m. The average coefficient of restitution of 12 mm HEDR layer is 0.6296.

The comparison of experimental and simulated results is shown in Fig. 6. It is clear that the simulated displacement and pounding force

agree well with the experimental results, which demonstrates the accuracy of the impact force model. Since the friction force between the striker and the VE materials layer is not considered in Eq. (10), the displacement and impact force are not accurate enough during the fourth pounding when the friction force may account for a huge part of the contact force.

3. Optimum design of SS-PTMD

Based on the validated impact force model, a numerical optimization based on the H_∞ optimization theory was performed to obtain the optimal parameters of the SS-PTMD, including optimal tuning frequency ratio and the optimal coefficient of restitution. The simplified design formulas for the SS-PTMD were given and further discussed.

3.1. Equations of motion of a SDOF system controlled by a SS-PTMD

As the mechanical model shown in Fig. 2, the equations of motion of a SDOF structure controlled by a SS-PTMD are given by,

$$\begin{cases} m_s \ddot{x}_s(t) + c_s \dot{x}_s(t) + k_s x_s(t) + k_d(x_s(t) - x_d(t)) = f(t) + F_p(t) \\ m_d \ddot{x}_d(t) + k_d(x_d(t) - x_s(t)) = -F_p(t) \end{cases} \quad (12)$$

where m_s , c_s , and k_s are the mass, damping coefficient and stiffness of the main structure, respectively. m_d and k_d are the mass and stiffness of the SS-PTMD. $x_s(t)$ and $x_d(t)$ are the displacement of the main structure and the SS-PTMD, respectively. f is the external loading and $F_p(t)$ is the pounding force. The pounding force $F_p(t)$ is equal to zero when the relative displacement $x_s(t) - x_d(t) < 0$. When the relative displacement $x_s(t) - x_d(t) > 0$, the pounding force $F_p(t)$ can be computed by Eq. (3). The indentation $\delta(t)$ in Eq. (3) should be substituted by,

$$\delta(t) = x_s(t) - x_d(t) \quad (13)$$

3.2. Numerical optimization based on H_∞ theory

The H_∞ optimization theory [54,55], which can minimize the maximum response of the structure in a wide frequency range, is considered in the numerical optimization. The optimization objective based on the H_∞ optimization theory is given by [48],

$$\min_{f_i, e_i} [\max_{\omega} (\text{DMF}(\omega))] \quad (14)$$

where $f_i = \omega_d/\omega_s$ represents the frequency tuning ratio of the SS-PTMD. ω_s is the circular frequency of the main structure and ω_d is the circular frequency of the SS-PTMD according to Eq. (1). e_i is the coefficient of restitution of the VE materials layer. DMF(ω), the dynamic magnification factor (DMF) of the main structure, is given by,

$$\text{DMF}(\omega) = A_s(\omega)k_s/f_0 \quad (15)$$

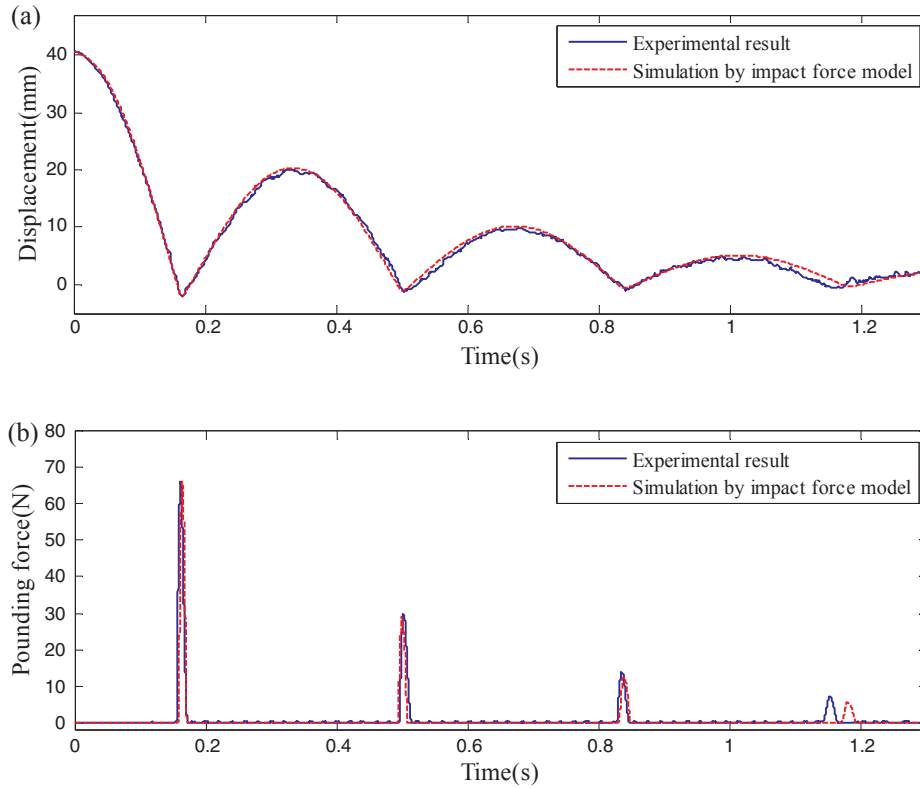


Fig. 6. Comparison of experimental and simulation results at an initial displacement 0.04 m: (a) Vertical displacement; (b) pounding force.

where $A_s(\omega)$ is the steady-state response amplitude of the main structure under a harmonic excitation force $f(t) = f_0 \sin(\omega t)$.

When a certain mass ratio of the SS-PTMD and the damping ratio of the main structure are given, the control performance of the SS-PTMD only depends on the frequency tuning ratio and the coefficient of restitution e of VE material layer. A range of frequency tuning ratio $f_i \in [0.94, 1.23]$ and the coefficient of restitution $e_i \in [0.3, 0.9]$ was specified in the numerical optimization. The SS-PTMD is usually designed to enhance the damping and reduce the large amplitude vibrations for low-damping structures, such as long-span bridges and high-risen buildings. These flexible structures have a low value of damping ratio, even less than 0.5–1%. Therefore, the damping ratio of the main structure is set to 0.5% in the optimization. The mass ratio of SS-PTMD is set from 0.01 to 0.05 with an increment of 0.01 and the excitation frequency ω of the harmonic loading $f(t) = f_0 \sin(\omega t)$ is varied from $0.8\omega_s$ to $1.2\omega_s$. Eq. (12) can be numerically solved by the Runge-Kutta fourth-order method to obtain the maximum DMF(ω) under different excitation frequencies, and the time interval of the solution is 1×10^{-4} s. Figs. 7 and 8 show the maximum DMF of the main structure when the frequency tuning ratio and the coefficient of restitution are varied in the search range. It is shown that the control performance of the SS-PTMD is insensitive to the derivation of the coefficient of restitution to the optimal value.

The optimal value of the frequency tuning ratio and the coefficient of restitution can be obtained by finding the point with the minimum value of the maximum DMF. The optimal frequency tuning ratio from the numerical optimization is plotted in Fig. 9(a) and the quadratic fitting curve is also shown. The optimal design formula for the frequency tuning ratio f_{opt} is given by,

$$f_{opt} = -28.6\mu^2 + 3.11\mu + 1.01 \quad (16)$$

where μ is the mass ratio of the SS-PTMD. It is shown from Fig. 8(a) that the optimal frequency tuning ratio for the SS-PTMD increases with the mass ratio. However for a conventional TMD system, the optimal

frequency tuning ratio decreases with the increase of the mass ratio. This is because the pounding duration will be enlarged due to the increase of the pounding mass, and consequently the frequency of the SS-PTMD decreases.

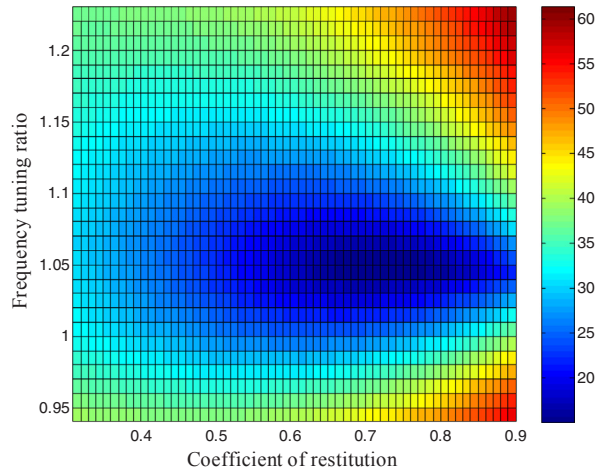
Fig. 9(b) shows the optimal coefficient of restitution e_{opt} from the numerical optimization, and the optimum design formula proposed by Wang et al. [48] is also plotted. It can be seen that the optimal coefficient of restitution from the optimization is slightly larger than the value from the formula. Considering the damping detuning of TMD system will not greatly affect the control performance, the optimum formula proposed by Wang et al. [48] is recommended in the design, and the formula is given by,

$$e_{opt} = \frac{1}{\exp\left(2\pi\sqrt{\frac{3\mu}{8(1+\mu)}}\right)} \quad (17)$$

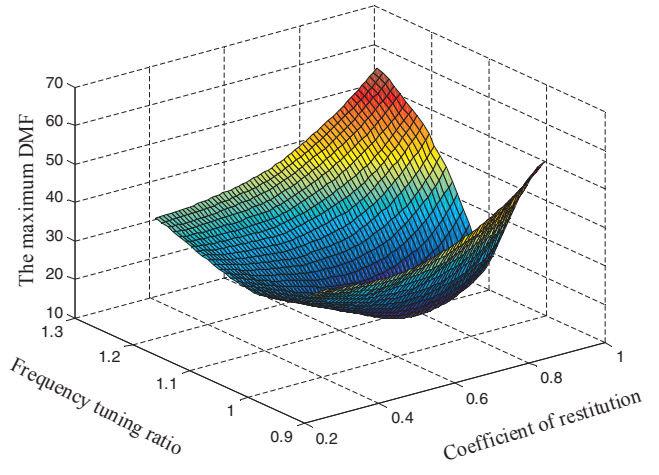
It should be noted that the wind loading such as the turbulence-induced buffeting force is generally a broad-band excitation which induces the multimode responses of a structure. For vortex-induced vibration in the present study, the excitation force is caused by the regular, periodic shedding of vortices in the wake of bridge decks. Although the nonlinear vortex shedding force is different to harmonic excitations, the vortex-induced vibration is generally dominated by a single mode [56], as also reflected from wind tunnel results of a bridge deck later. Therefore, the optimal parameters of the SS-PTMD obtained by applying a harmonic excitation on the primary structure are applicable to control the structural response under VIVs.

4. Control of a bridge deck model by a SS-PTMD under VIV

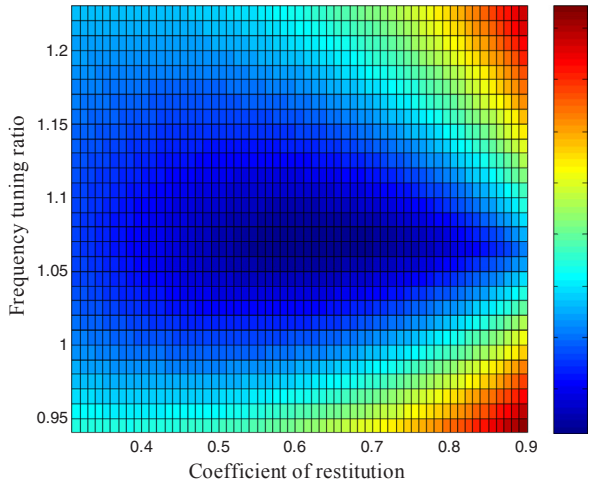
A sectional model test in wind tunnel is the most convenient way to study the various wind-induced responses, such as flutters and VIVs, of actual flexible bridges subjected to wind loadings. In the sectional model test, the dynamic properties of a flexible bridge are first determined from eigenvalue analysis, and the sectional model of the



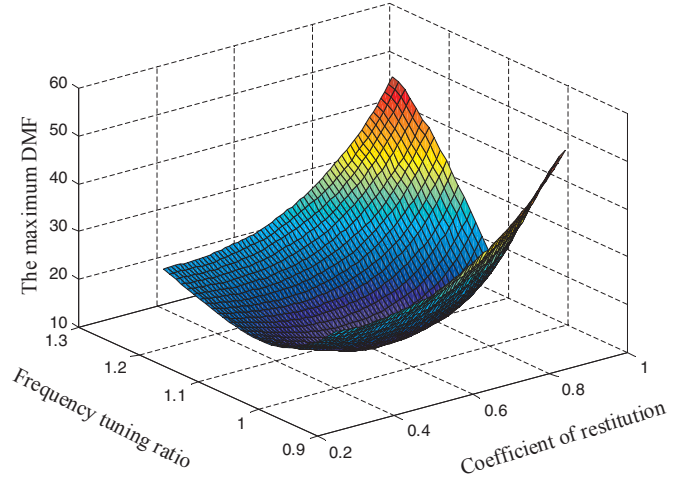
(a) Top view



(b) 3D view

Fig. 7. Maximum DMF of main structure for $\mu = 0.01$: (a) Top view; (b) 3D view.

(a) Top view



(b) 3D view.

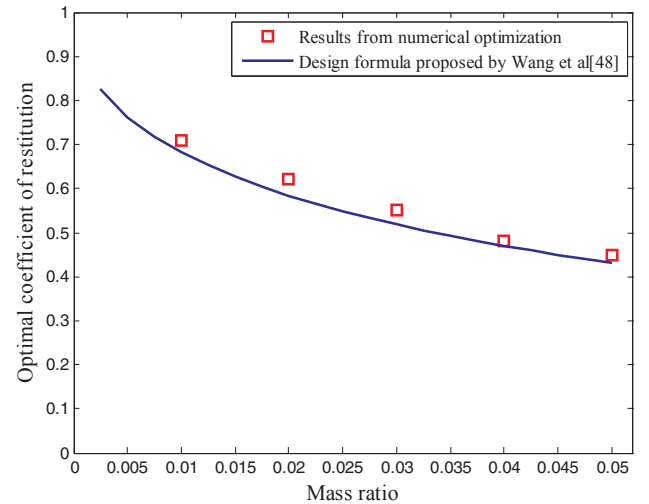
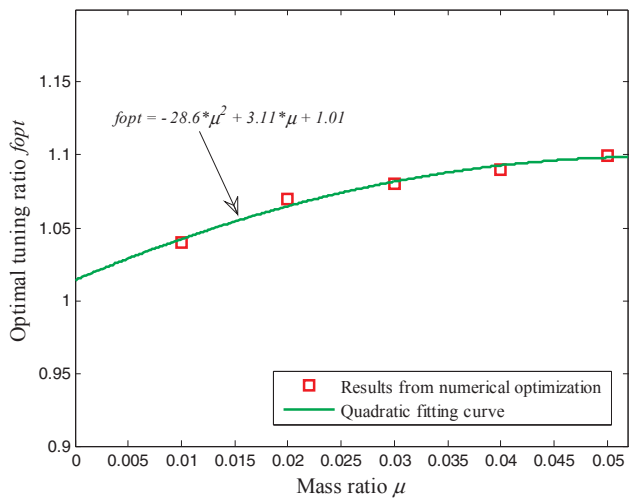
Fig. 8. Maximum DMF of main structure for $\mu = 0.02$: (a) Top view; (b) 3D view.

Fig. 9. Optimal parameters of SS-PTMD: (a) Frequency tuning ratio; (b) coefficient of restitution.

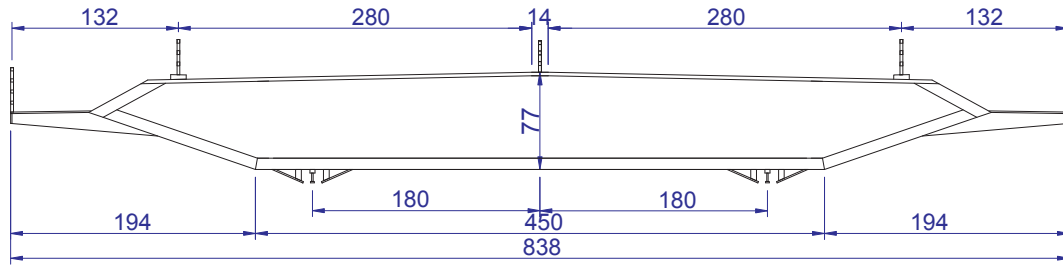


Fig. 10. Cross section of the tested bridge deck model (mm).

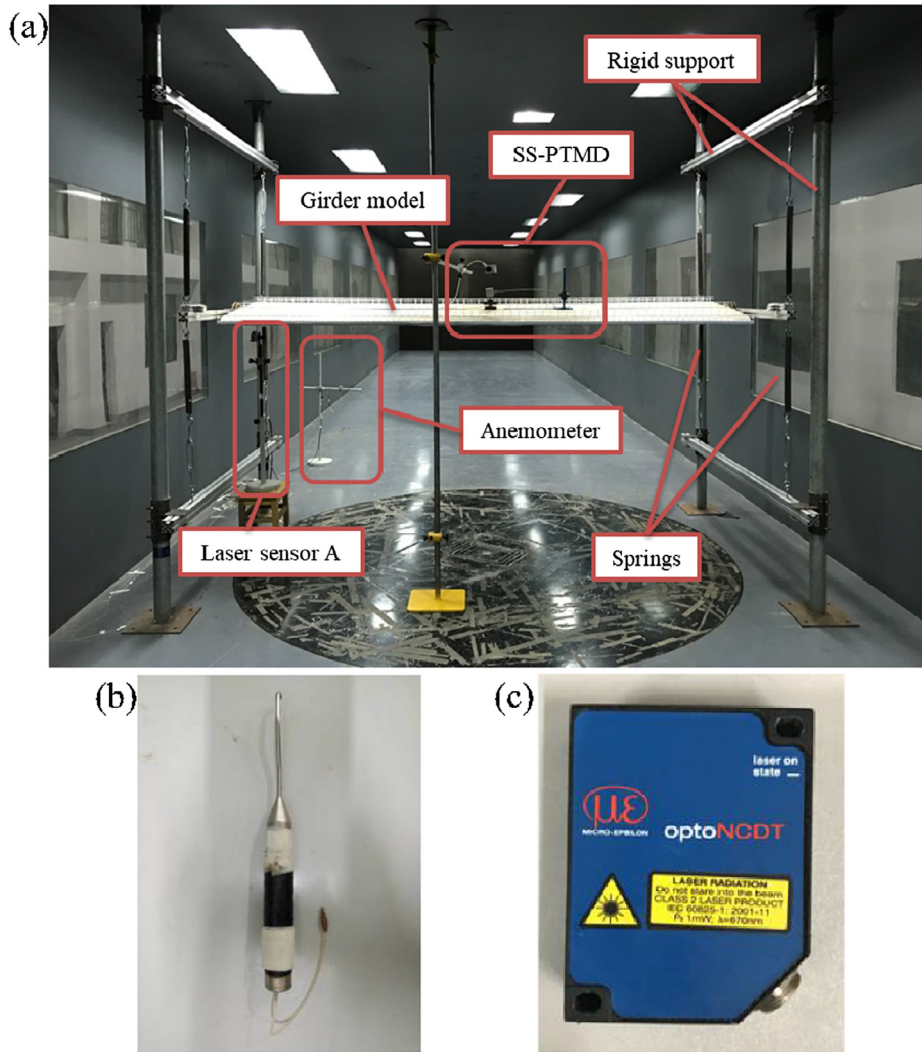


Fig. 11. Experimental setup of VIV tests in a wind tunnel: (a) Overview; (b) Cobra Probe; (c) laser displacement sensor.

Table 1
Dynamic parameters of bridge deck sectional model.

Mass m_m (kg)	Stiffness k_m (N/m)	Frequency f_m (Hz)	Damping coefficient c_m (N·m/s)	Damping ratio ζ_m
43.2	14343.0	2.90	11.02	0.70%

bridge deck is designed based on the modal properties of fundamental vertical mode and torsional mode. The sectional model is elastically mounted by coil springs in wind tunnel to simulate the target modal properties, and is tested in wind tunnel at varying wind speeds. Therefore, only a bridge deck model is used to numerically and

experimentally study the effectiveness of a SS-PTMD for control of vortex-induced vibrations. The emphasis has been placed on considering nonlinear features of vortex shedding forces.

4.1. Modeling of a bridge deck model with a SS-PTMD under VIV

The vortex shedding force differs from the sinusoidal force due to its nonlinear feature of fluid and dependence on vibration amplitude. A semi-empirical model [57], based on van der Pol oscillator, was used to model the vortex-shedding vibration [57,58] in this section. Consider a section model with a height D and length L . According to Facchinetti et al. [57], the motion equations of elastically mounted bridge deck model immersed in flow is given by,

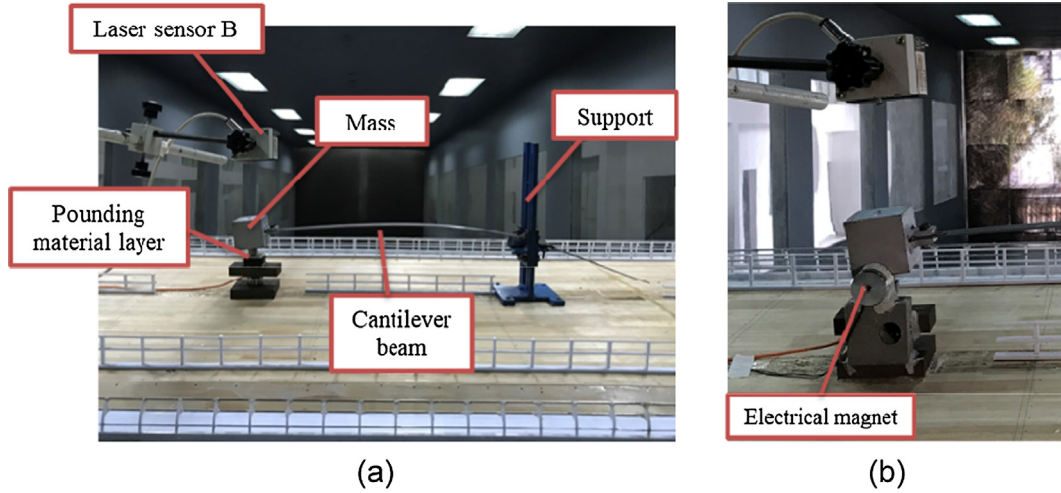


Fig. 12. Details of wind tunnel test: (a) SS-PTMD device; (b) locking device using an electrical magnet.

Table 2
Dynamic parameters of SS-PTMD.

Case	Mass m_d (kg)	Mass ratio	Stiffness k_d (N/m)	Frequency f_d (Hz)	Frequency tuning ratio	COR e
Optimal value	0.86	0.02	80.36	3.077	1.061	0.5835
Experimental value			81.57	3.100	1.069	0.6296

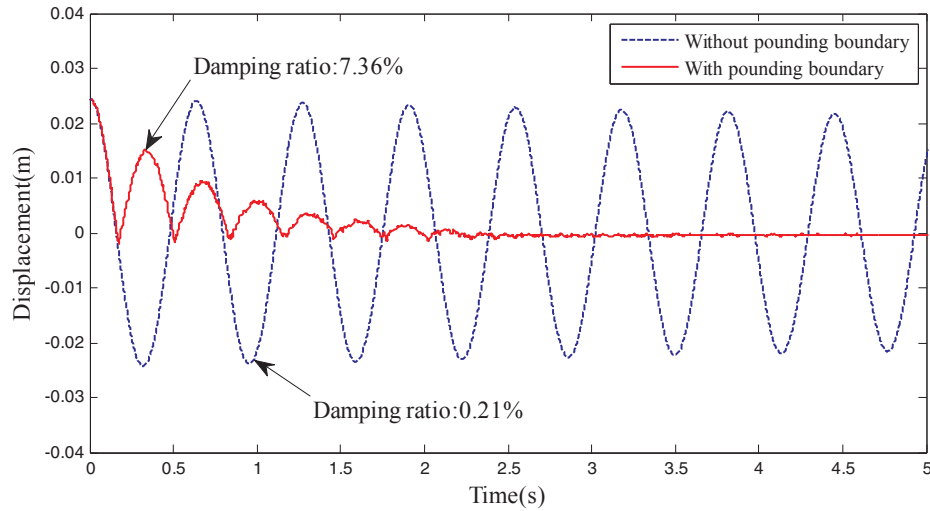


Fig. 13. Free vibration responses of the SS-PTMD with and without HEDR pounding boundary.

$$m_m \ddot{x}_m(t) + c_m \dot{x}_m(t) + k_m x_m(t) = F_L \quad (18)$$

$$F_L = (\gamma \omega_f \rho D^2 L) \dot{x}_m(t) + \left(\frac{1}{4} \rho U^2 D C_{L0} L \right) q(t) \quad (19)$$

$$\ddot{q}(t) + \varepsilon \omega_f [q^2(t) - 1] \dot{q}(t) + \omega_f^2 q = [\omega_f (A/D)] \dot{x}_m(t) \quad (20)$$

where m_m , c_m and k_m are the mass, damping coefficient and stiffness of the spring-mounted bridge deck, respectively. $x_m(t)$ and $q(t)$ are the crosswind response of the model and the wake variable. F_L represents the nonlinear vortex shedding lift force based on wake oscillator model. ω_f is the frequency of vortex shedding, which is linear proportional to flow velocity outside lock-in range. ρ is the air density. γ is a stall parameter and C_{L0} is the magnitude of the lift coefficient caused by vortex. A is the velocity coupling coefficient and ε is the van der Pol coefficient. The four unknown model parameters (γ , C_{L0} , A , ε) in the wake oscillator model can be identified from vortex-induced vibrations of the cross section model in a wind tunnel test, as will be shown later.

When the bridge deck model is controlled by the SS-PTMD, the coupling system of the deck model and wake oscillator expressed by Eqs. (18) and (20) are given as,

$$m_m \ddot{x}_m(t) + c_m \dot{x}_m(t) + k_m x_m(t) = F_L + F_p(t) + k_d(x_d(t) - x_m(t)) \quad (21)$$

$$\ddot{q}(t) + \varepsilon \omega_f [q^2(t) - 1] \dot{q}(t) + \omega_f^2 q = [\omega_f (A/D)] \dot{x}_m(t) \quad (22)$$

$$m_d \ddot{x}_d(t) + k_d(x_d(t) - x_m(t)) = -F_p(t) \quad (23)$$

where Eqs.(21)–(23) represent the motion equations of the deck model, the wake oscillator and the SS-PTMD. $F_p(t)$ is the pounding force, and it can be computed by Eq. (3) when the pounding condition is satisfied. The procedures of the proposed simulation method to compute the dynamic response of the bridge deck model are summarized below:

- (1). Measure the VIV response of the bridge deck model which is elastically supported in a wind tunnel under different wind speeds.

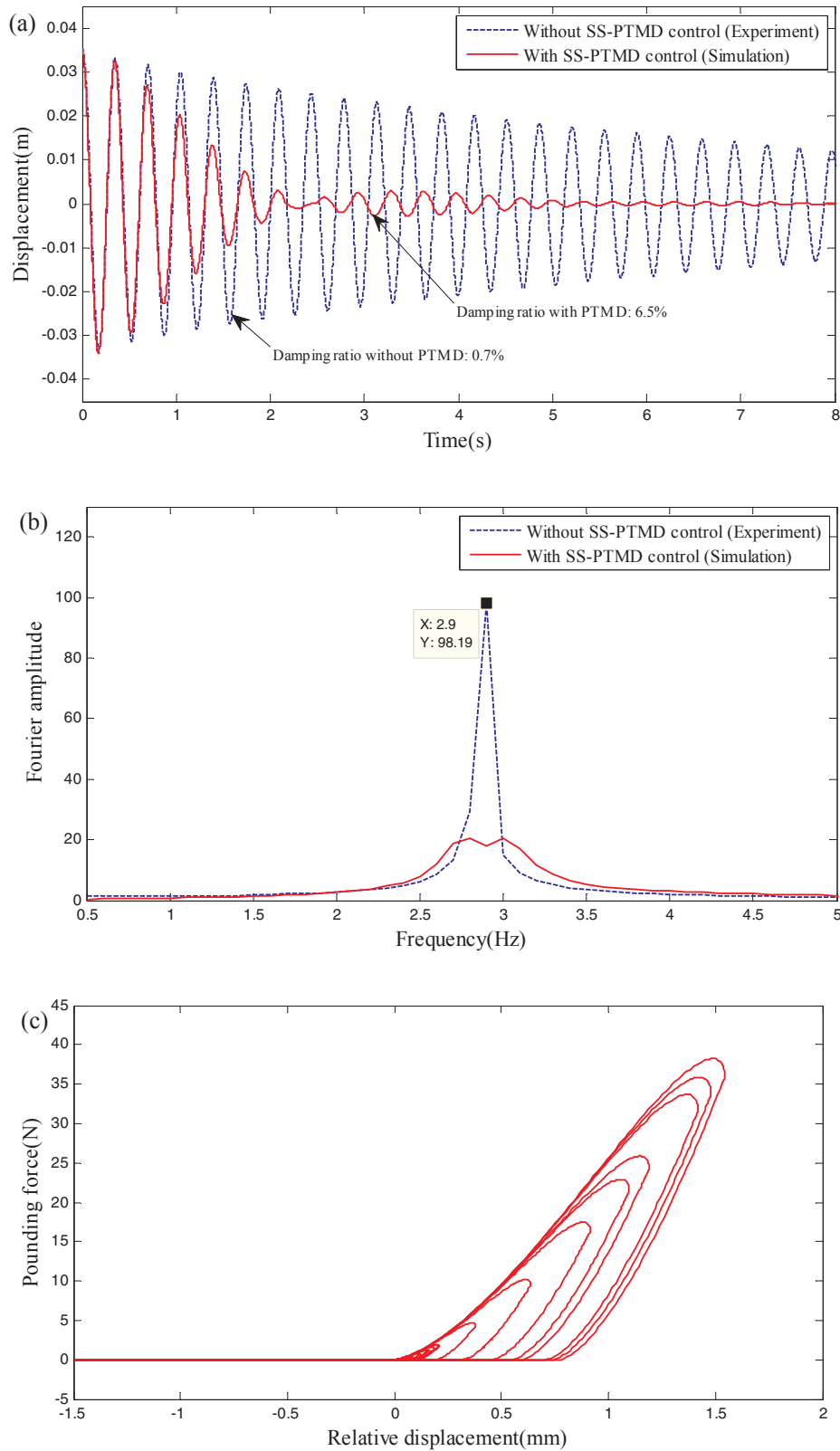


Fig. 14. Free vibration response of deck sectional model: (a) Time history; (b) amplitude-frequency response; (c) pounding force-relative displacement loop; (d) spring force-relative displacement loop.

(2). Use the Runge-Kutta fourth order method to compute the displacement of the bridge deck model when the unknown parameters (γ , C_{L0} , A , ϵ) in Eqs. (18)–(20) is assumed to certain values. Due to the existence of the negative damping, the VIV displacement of the bridge deck model will eventually converge to a stable-state

response even if a random initial condition is given to Eqs. (18) and (20). Then a wind speed-VIV amplitude curve of the bridge deck can be obtained from the numerical simulation. The optimal value of the four unknown parameters (γ , C_{L0} , A , ϵ) to describe vortex shedding lift force should be identified through a numerical

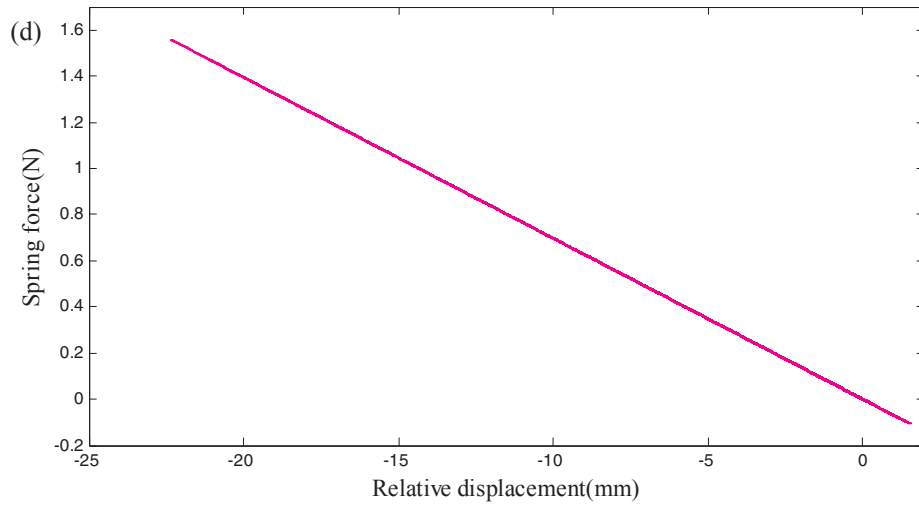


Fig. 14. (continued)

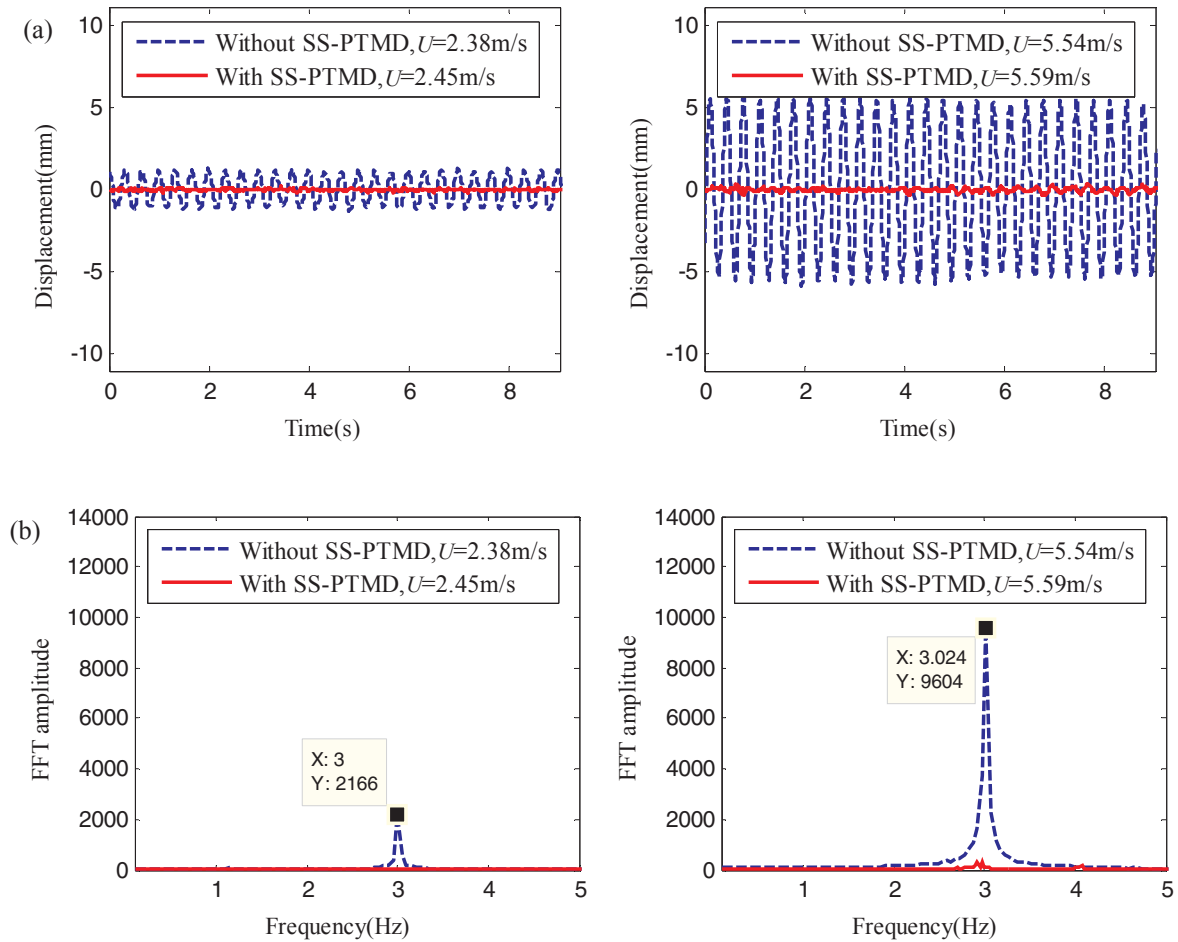


Fig. 15. Displacement of deck model and the corresponding frequency responses under different wind speeds: (a) Time history; (b) frequency response.

optimization program, letting the normalized error of the wind speed-VIV amplitude curve from simulation and wind tunnel tests to be minimized.

- (3). Substitute the identified model parameters and the parameters of the SS-PTMD into Eqs. (21)–(23) and compute the wind speed-displacement amplitude curve of the bridge deck model with a SS-PTMD under VIV.

4.2. Experimental setup and dynamic properties of the model

A streamlined cross section shown in Fig. 10 was selected to study the VIV and the control effectiveness of the SS-PTMD for VIV control. It should be noted that the cross section of bridge deck studied in this paper is a typical one, and similar cross section has been adopted in some long span bridges such as the Tsurumi Bridge [59] and the Great Belt East Bridge [60]. As vortex-induced vibration is characterized by a single mode, one single vertical mode is simulated in wind tunnel test.

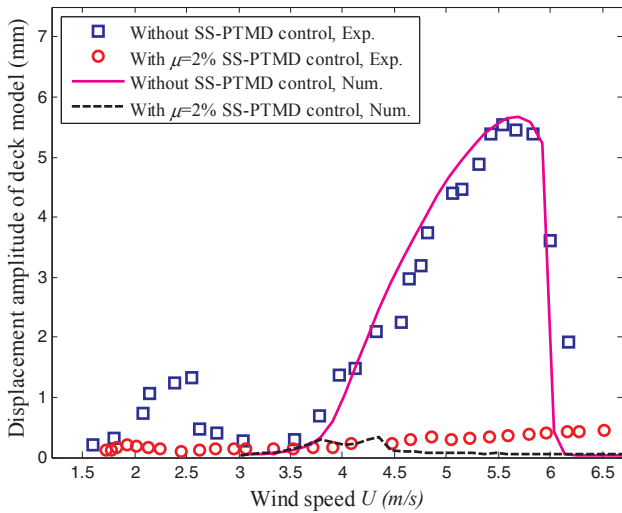


Fig. 16. Displacement amplitude of deck model under different wind speeds.

Table 3

Identified parameters for wake oscillator model.

Parameter	Stall parameter γ	Magnitude of lift coefficient C_{L0}	Velocity coupling coefficient A	van der Pol coefficient ε
Value	6.949	0.181	38.8	0.072

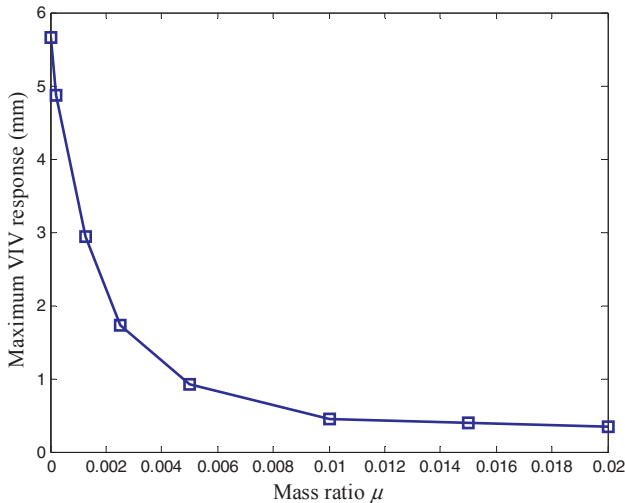


Fig. 17. Maximum VIV response of the deck model with a SS-PTMD at different mass ratios.

Therefore, both in numerical and experimental study, a scaled-down bridge deck model elastically supported by springs, which can capture the natural frequency of one vibration mode of the basic structure, are used to consider the dynamic behavior of bridge deck under VIVs.

The experimental setup of the wind tunnel test is shown in Fig. 11 and Table 1 shows the dynamic properties of the section model. The wind tunnel test was conducted in the KD-1 Wind Tunnel at Hunan University of Science and Technology. The deck model with a cross-flow length of $L = 3.0$ m was suspended by 8 linear springs in the wind tunnel and the effective mass of the bridge deck model is $m_m = 43.2$ kg. The laser displacement sensor A (model: optoNCDT14, which was manufactured by $\mu\epsilon$ Company) was used to measure the vertical displacement of the model, and a pressure anemometer (model: TFI-Series 100 Cobra Probe, which was manufactured by Turbulent Flow Instrumentation Company) was placed in the front of the model to

collect the wind speed during the wind tunnel test.

4.3. Design of SS-PTMD

A SS-PTMD device was designed to place on the deck model to suppress the response of the deck model under VIV, as shown in Fig. 12(a). The frequency of the SS-PTMD can be adjusted by changing the length of the cantilever beam and the equivalent damping ratio depends on the coefficient of restitution of the used VE materials layer. The laser displacement sensor B (model: optoNCDT14, manufactured by $\mu\epsilon$ Company) was used to record the displacement of the SS-PTMD. The existence of the SS-PTMD device may affect the aerodynamic property of the bridge deck model. To keep the consistence on the aerodynamic property of the tested model under controlled and uncontrolled case, a locking device using an electrical magnet, as shown in Fig. 12(b), was employed to lock and release the SS-PTMD during the wind tunnel test. When the uncontrolled case is tested, a 20 V DC power is applied to the magnet and the SS-PTMD will move together with the model due to magnetic force. When the power supply for the magnet is cut down, the SS-PTMD will be released to absorb the vibration energy from the deck model and the vibration energy will be dissipated by poundings.

The mass of the SS-PTMD installed on the deck model is 0.86 kg and consequently the mass ratio of the SS-PTMD is $\mu = 0.02$. According to Eqs. (16) and (17), the optimal frequency tuning ratio is 1.061 and the optimal coefficient of restitution of the VE material layer is 0.5835. From Eq. (1), the optimal stiffness provided by the cantilever beam to achieve the optimal frequency 3.077 Hz should be 80.36 N/m. When the length of the cantilever beam is set to 45.6 cm, the stiffness provided by the beam is 81.57 N/m and consequently the frequency of the SS-PTMD is 3.1 Hz according to Eq. (1). It should be noted that the stiffness provided by the cantilever beam can be calculated from the free vibration tests of the SS-PTMD when the pounding boundary is removed. Since the effectiveness is insensitive to the detuning of the coefficient of restitution, the 12 mm HEDR layer with a coefficient of restitution $e = 0.6296$, was used as the pounding boundary. The optimal and experimental parameters of the SS-PTMD are summarized in Table 2. It can be seen from Figs. 7 and 8, the deviation of the restitution coefficient to the optimal value only have litter influence on the performance. Although the coefficient of restitution of the HEDR pounding boundary is not quite close to the optimal value, the employed SS-PTMD device still can be regarded as an optimized one.

Fig. 13 shows the free vibration responses of the designed SS-PTMD when the 12 mm HEDR pounding boundary is installed and uninstalled. It is evident from Fig. 13 that the period of the case with the pounding boundary becomes half to that without the boundary and the damping ratio of the SS-PTMD is significantly enhanced from 0.21% to 7.36% by poundings.

With the parameters listed in Tables 1 and 2, the free vibration response of the bridge deck model with SS-PTMD is obtained by numerically solving Eq. (12) and the result is plotted in Fig. 14(a) and (b) to compare with the uncontrolled result obtained from the experiment. It can be seen that the damping ratio of the model increases from 0.70% to 6.5% after the installation of the SS-PTMD, and the amplitude-frequency curve also indicates that the vibration amplitude is greatly suppressed. Fig. 14(c) plots the hysteretic loop between the pounding force and the relative displacement of the deck model and SS-PTMD. It can be seen that the energy dissipation of the deck-SS-PTMD system only happens when the relative displacement is greater than zero (or when poundings between the deck and SS-PTMD are occurred). Fig. 14(d) plots the hysteretic loop between the spring force and the relative displacement of the deck model and SS-PTMD. It is shown that the spring force does not dissipate the vibration energy at all.

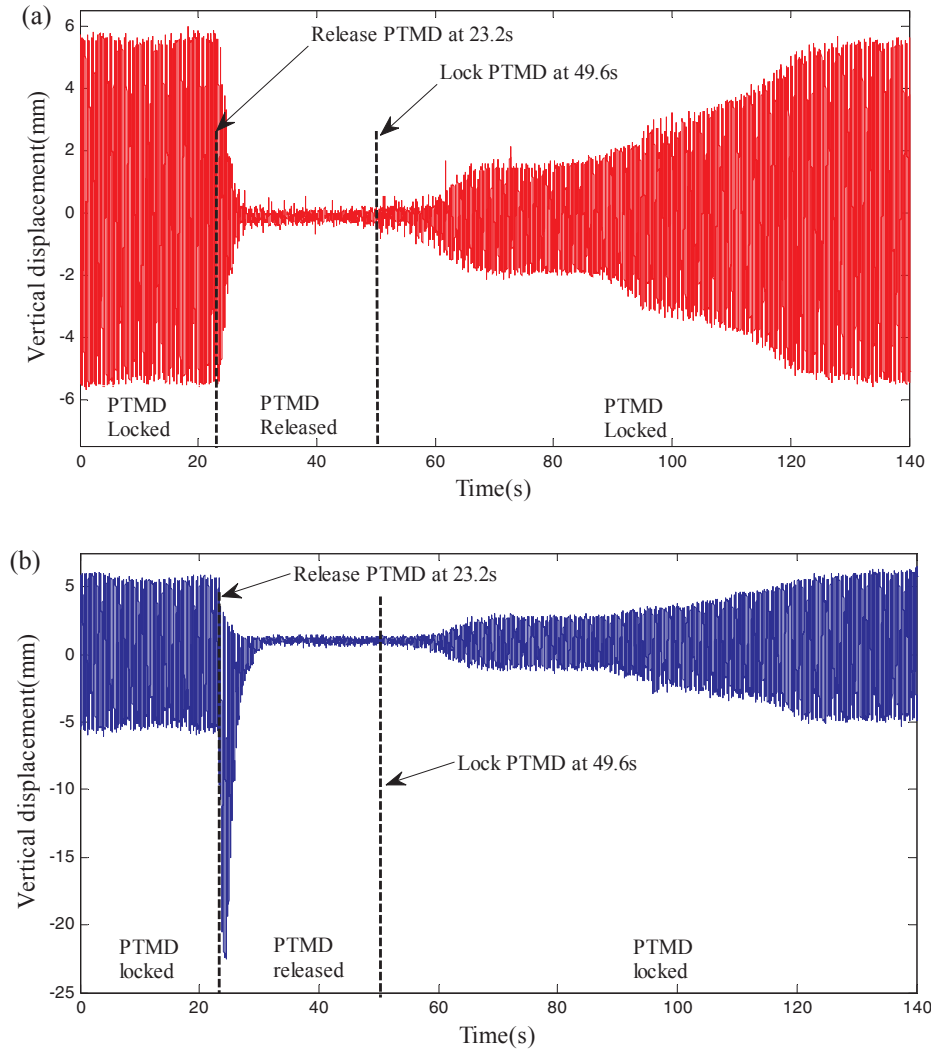


Fig. 18. Lock-release experiment under wind velocity $U = 5.54$ m/s: (a) Response of the bridge deck model; (b) response of SS-PTMD.

4.4. Wind tunnel results

The sectional model test in the wind tunnel was carried out to study the effectiveness of SS-PTMD for VIV control. The displacement histories of the deck model and the corresponding frequency response in the wind speed around 2.4 m/s, 4.6 m/s, 5.5 m/s and 6.0 m/s are shown in Fig. 15. It is shown that the vibration frequencies of the bridge deck model in the uncontrolled case under VIV are slightly larger than its natural frequency measured in the static air. The displacement amplitude of the bridge deck model without and with the optimized SS-PTMD in the wind speed range from 1.5 m/s to 6.5 m/s are shown in Fig. 16. It can be seen from the uncontrolled case that there are two lock-in regions of VIVs, which are $U = [1.8 \text{ m/s}, 2.8 \text{ m/s}]$ and $U = [3.8 \text{ m/s}, 6.5 \text{ m/s}]$, respectively. The maximum amplitude of VIV of 5.56 mm occurred in the second lock-in region at the wind speed of 5.54 m/s. When the SS-PTMD was activated, the large amplitude VIV of the model disappeared and the maximum VIV response of the model is suppressed by 94% compared to the uncontrolled case at the wind speed of 5.54 m/s. It is also shown that the large amplitude VIV responses in both the two lock-in regions are controlled.

4.5. Numerical results

The second VIV region $U = [3.8 \text{ m/s}, 6.5 \text{ m/s}]$ from the experimental result of the uncontrolled case, as plotted in Fig. 16, were

utilized to identify the four unknown parameters (γ , C_{LOA} , ϵ) in the wake oscillator model. Eqs. (18) and (19) can be numerically solved by the fourth order Runge-Kutta method to obtain the VIV responses under different wind speeds. The optimal values of the unidentified parameters were obtained by minimizing the normalized error between experimental and numerical results. The expression of the normalized error is given by,

$$err = \sum_{i=1}^n \sqrt{(E_i - N_i)^2} \quad (24)$$

where E_i and N_i are respectively the experimental and the numerical values of the vortex-induced response amplitude at the i th wind speed. An initial condition for the numerical solution of Runge-Kutta method is set to $[x_{m0} = dis_{i-1}, \dot{x}_{m0} = 0, q_0 = 0, \dot{q}_0 = 0]$, where dis_{i-1} is the steady-state vibration amplitude of the deck model from the last time computation [57]. The time step of the solution is 1×10^{-4} s and the wind speeds used in the numerical solution was set as same as that measured from wind tunnel experiments. The optimal values of the parameters are listed in Table 3 through the numerical identification program. The uncontrolled case from the simulation using the identified parameters is plotted in Fig. 16. It can be seen that the numerical results can well reflect the VIV response of the deck model when the SS-PTMD is deactivated. The identified parameters summarized in Table 3 will be continuously used to analyze the case with SS-PTMD control. In this paper, the four unknown parameters in the wake oscillator model

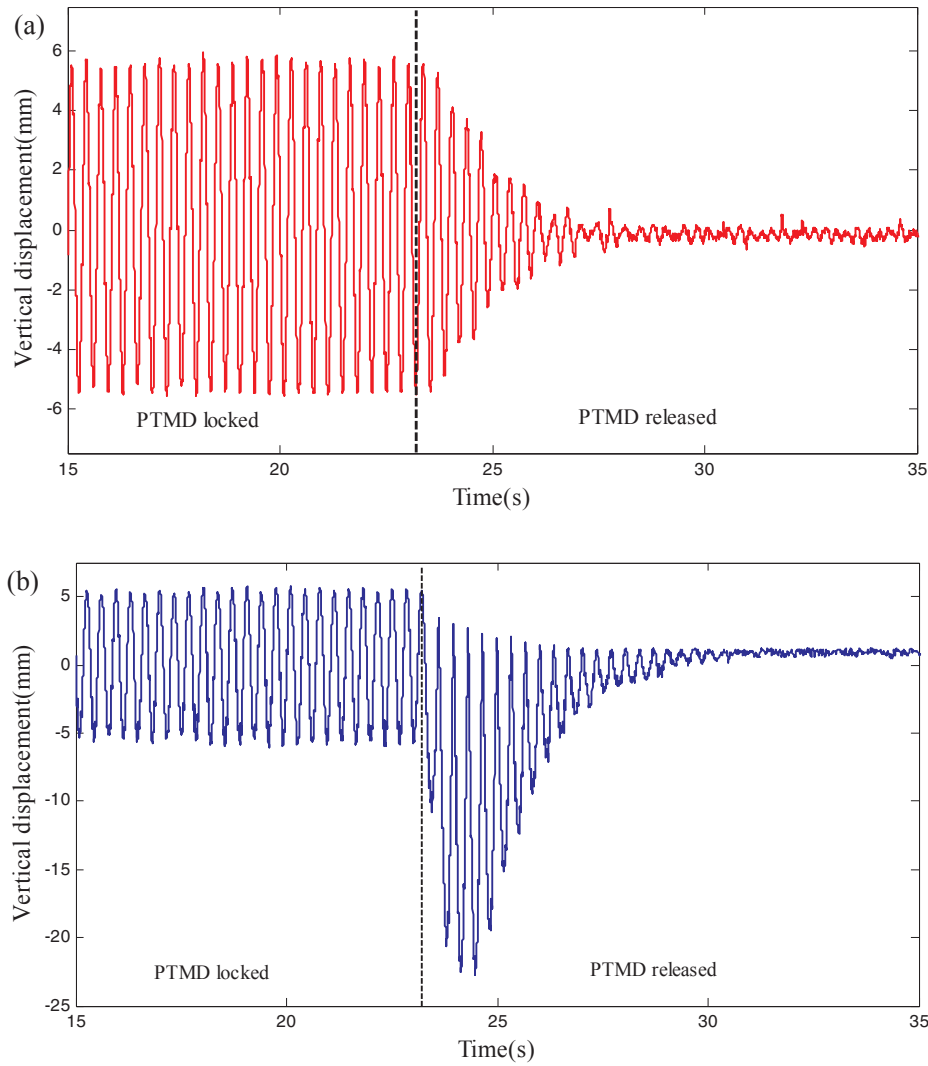


Fig. 19. Details of lock-release experiment from 15 s to 35 s: (a) Response of the bridge deck model; (b) response of SS-PTMD.

are identified from the numerical identification method following the previous studies on VIVs of rectangular section model [61]. Actually, the vortex lift coefficient can be obtained from wind tunnel tests by fixing the structure subjected to vortex shedding and measure the lift force. However due to the nonlinear feature of VIVs, the vortex lift force as well as C_{L0} are dependent on the vibration amplitude of the bridge deck under VIVs. Therefore C_{L0} is also taken as an unknown parameter in the identification.

The VIV response of the deck model controlled by a SS-PTMD with mass ratio of 2% was numerically computed by Eqs. (21)–(23) and the numerical results are also shown in Fig. 16 to compare with experimental results. It is shown that the numerical result matches well with the experimental data at the lower wind speed of the lock-in region, and however the error between experimental and numerical results at the higher wind speed of the lock-in region become greater and greater. This is because the van der Pol oscillator model only can take the self-excited vibration into account and the buffeting response of the deck model increases with the increase of the wind speed during wind tunnel tests.

Moreover, the deck model controlled by the SS-PTMD with different mass ratios was computed. Fig. 17 plots the maximum VIV response of the deck model controlled by the SS-PTMD with various mass ratios. It is shown that at the lower value of mass ratio such as less than 0.5%, the maximum response dramatically reduces with mass ratio, and however the performance of the SS-PTMD will not be enhanced too

much with the increase of the mass ratio when the mass ratio is greater than 1%.

4.6. A lock-release experiment under the maximum VIV

A lock-release experiment was carried out to reveal the dynamic characteristics of the SS-PTMD to control the large amplitude VIV under wind speed $U = 5.54$ m/s, where the maximum vortex-induced response was occurred. The vertical responses of the model and the SS-PTMD during the lock-release experiment are shown in Fig. 18. It is shown that the large amplitude VIV occurred during the first 23.2 s when the SS-PTMD was locked by the electromagnet. After that the SS-PTMD was released by cutting off the DC power to the electromagnet, and the vibration amplitude rapidly decreased from 5.56 mm to 0.33 mm with a reduction of 94%. After locking the SS-PTMD at 49.6 s, the large amplitude VIV occurred again and the response amplitude returned to the value as same as that during the first 23.2 s. Moreover, the displacement of the SS-PTMD suddenly increased from the released moment, and the vibration energy was dissipated by the poundings between the SS-PTMD and the HEDR layer in the next few seconds. The zoomed view of the lock-release experiment between 15 s and 35 s is shown in Fig. 19. It should be noted that the lock-release experiment not only strongly demonstrates the effectiveness of the SS-PTMD for suppression of VIVs, but also reveals the dynamic characteristics of the SS-PTMD when is released.

5. Conclusions

This paper proposes a new method to control the vortex-induced vibration (VIV) of a bridge deck model by using singled-sided PTMD (SS-PTMD) with one VE material pounding boundary. An impact force model with the consideration of the poundings between VE materials layer and steel was employed to simulate the dynamic response of the SS-PTMD and was then validated by free pounding experiments. The coefficient of restitution of different VE material layers was also studied. Simplified design formulas for the PTMD were obtained from a numerical optimization based on the verified impact force model. The effectiveness of the proposed SS-PTMD was studied by wind tunnel tests and numerical simulations.

The coefficient of restitution of VE material layer is stable under various impact velocities and can be adjusted in a certain range by using a composite layer of VE materials. The results from wind tunnel tests show that there are two lock-in regions appeared when the PTMD is deactivated. The designed PTMD with mass ratio of 2% can reduce the maximum amplitude of VIVs by 94%. The performance of the PTMD may not be enhanced too much with the increase of the mass ratio, when mass ratio of the PTMD is larger than 1%. The lock-release experiment can well demonstrate the effectiveness of the PTMD to control VIVs resulted from wind excitations.

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